

# **Exponential and Log Functions**

**ECON 441: Introduction to Mathematical Economics**  
**Lecture 10**

Div Bhagia

## Example

Total cost:  $C = C(Q)$

Marginal cost:  $MC = C'(Q)$

Average cost:

$$AC = \frac{C(Q)}{Q}$$

When is  $\frac{dAC}{dQ}$  positive?

## Example

Revenue:  $R = f(Q)$

Output:  $Q = g(L)$

How does revenue change due to a change in labor input  $L$ ?

$$\frac{dR}{dL} = \frac{dR}{dQ} \cdot \frac{dQ}{dL}$$

# Exponential Functions

The exponential function can be represented as:

$$y = f(x) = b^x \quad (b > 1)$$

where  $b$  denotes a fixed base of the exponent.

A more generalized version can be written as:

$$y = ab^{cx}$$

# Natural Exponential Function

**Natural exponential function:** the base is a special mathematical constant called Euler's number  $e = 2.71828\dots$

$$y = ae^{rx}$$

Where did this number  $e$  come from?

It can be shown:

$$e \equiv \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

# Natural Exponential Function

Jacob Bernoulli discovered this constant in 1683 while studying a question about compound interest.

Invest  $A$  dollars at a 5% interest rate that compounds yearly. After one year you have  $A(1 + 0.05)$ ; after  $t$  years:

$$A(1 + 0.05)^t$$

# Natural Exponential Function

After  $t$  years:  $A(1 + 0.05)^t$ . Rewrite this as:

$$A \left( \underbrace{\left( 1 + \frac{1}{1/0.05} \right)^{1/0.05}}_{\approx e} \right)^{0.05t} \approx Ae^{0.05t}$$

More generally, with initial principal  $A_0$ , interest rate  $r$ , and horizon  $t$ :

$$A_t \approx A_0 e^{rt}$$

# Logarithmic Function

Since the exponential function is a monotonic function, its inverse exists.

The inverse of the exponential function is called the **log** or **logarithmic** function.

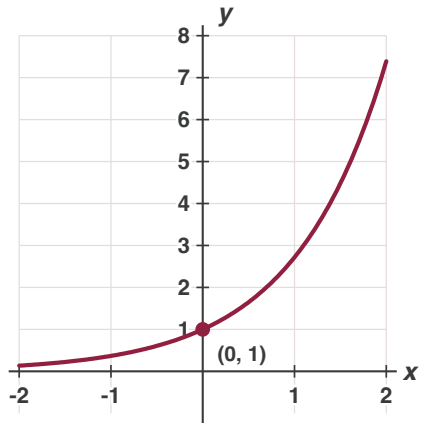
For the exponential function:

$$y = b^x \rightarrow \log_b y = x$$

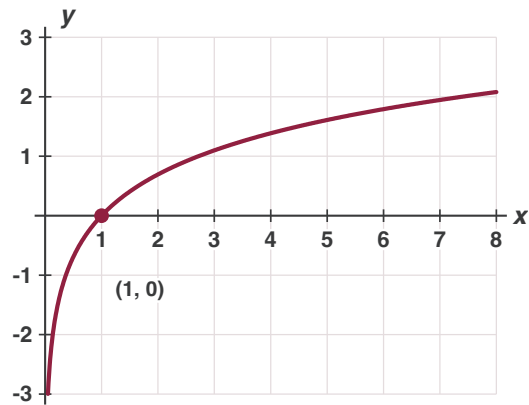
For the natural exponential function:

$$y = e^x \rightarrow \log_e y = \ln y = x$$

$$y = e^x$$



$$y = \ln x$$



# Rules for Logarithmic Functions

Let  $p = \ln u$  and  $q = \ln v$ , so that  $u = e^p$  and  $v = e^q$ .

$\ln(uv) = \ln u + \ln v$ , since

$$uv = e^p e^q = e^{p+q} \Rightarrow \ln(uv) = p + q$$

$\ln(u/v) = \ln u - \ln v$ , since

$$\frac{u}{v} = \frac{e^p}{e^q} = e^{p-q} \Rightarrow \ln\left(\frac{u}{v}\right) = p - q$$

$\ln u^a = a \ln u$ , since

$$u^a = (e^p)^a = e^{ap} \Rightarrow \ln u^a = ap$$

# Derivatives of Exponential Functions

Derivative of the exponential function:

$$y = e^x \quad \rightarrow \quad \frac{dy}{dx} = e^x$$

Using the chain rule:

$$y = e^{f(x)} \quad \rightarrow \quad \frac{dy}{dx} = f'(x)e^{f(x)}$$

# Derivatives of Logarithmic Functions

Derivative of the log function:

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

Using the chain rule:

$$\frac{d}{dx} \ln f(x) = \frac{f'(x)}{f(x)}$$

# Examples

Find the derivatives for the following functions:

1.  $y = e^x$

2.  $y = \ln x$

3.  $y = ae^{rx}$

4.  $y = e^{-x}$

5.  $y = \ln ax$

6.  $y = \ln x^c$

# Homework Questions

- Exercise 10.5: 1, 3, 7

Textbook reference: 10.5

Notes on Log and Exponential Functions are on the website